60-Day Python Roadmap for MLOps + GitHub Portfolio

This roadmap is optimized for:

- practical Python fluency
- MLOps preparation
- building a visible GitHub portfolio
- avoiding tutorial paralysis

Structure:

- **2–3 hours/day**
- every day includes:
    - learning
    - coding
    - GitHub documentation

Daily Non-Negotiables

Every day:

1. Push to GitHub

Even if tiny.

2. Write a short learning note

Example:

Today I learned:
- loops
- list comprehensions
- why input() returns strings

3. Write code manually

Do not only watch videos.

Recommended Repo Structure

```
python-mlops-journey/
│
├── week-01/
├── week-02/
├── projects/
├── notes/
├── exercises/
└── README.md
```

PHASE 1 — Python Foundations (Days 1–14)

Goal:
Become comfortable writing small Python programs.

Day 1

Learn

- Python installation
- VS Code
- running Python files
- print()
- variables

Build

```
name = "John"
print(name)
```

GitHub

Create your main repo.

Day 2

Learn

- data types
- type()
- strings
- numbers
- booleans

Exercises

- temperature converter
- age calculator

Day 3

Learn

- input()
- type conversion
- f-strings

These are fundamental beginner concepts.

Build

Interactive greeting app.

Day 4

Learn

- if
- elif
- else
- comparison operators

Build

Even/odd checker.

Practice like this is highly recommended early.

Day 5

Learn

- loops
- range()
- while loops

Build

Multiplication table generator.

Day 6

Learn

- lists
- indexing
- append()
- remove()

Build

Shopping list app.

Day 7

Mini Project #1

Build

CLI Calculator

Features:

- add
- subtract
- multiply
- divide
- error handling

Mini projects accelerate learning dramatically.

Day 8

Learn

- dictionaries
- key/value pairs
- .get()
- looping dictionaries

Build

Student record manager.

Day 9

Learn

- sets
- tuples
- uniqueness
- membership checking

Build

Duplicate remover.

Day 10

Learn

- functions
- parameters
- return

Functions are core Python building blocks.

Build

Calculator refactor using functions.

Day 11

Learn

- scope
- imports
- modules

Build

Random password generator.

Day 12

Learn

- error handling
- try/except
- ValueError

Debugging skills matter early.

Build

Safe number parser.

Day 13

Learn

- file reading
- file writing
- with open()

Build

Simple notes saver.

Day 14

Review Day

Refactor everything.

Focus on:

- readability
- naming
- formatting

Pythonic readability is extremely important.

PHASE 2 — Intermediate Python (Days 15–30)

Goal:
Write useful scripts and understand structured data.

Day 15

Learn

- list comprehensions

Build

Square-number generator.

Day 16

Learn

- enumerate()
- zip()
- useful built-ins

Exercises

Practice loops and iteration.

Day 17

Learn

- string methods
- split()
- join()
- strip()

Build

Word frequency counter.

Day 18

Learn

- JSON basics

Build

Save/load user data in JSON.

Day 19

Learn

- CSV files
- csv module

Build

CSV reader script.

Day 20

Mini Project #2

Build

Expense Tracker

Features:

- add expenses

- save CSV
- summarize totals

Excellent intermediate project.

Day 21

Learn

- pathlib
- os

Build

Directory scanner.

Day 22

Learn

- virtual environments
- pip

Practice

Install:

- requests
- pandas

Day 23

Learn

- APIs
- requests

Build

Weather API app.

Day 24

Learn

- exceptions deeper
- custom messages

Refactor

Improve previous projects.

Day 25

Learn

- argparse

Build

Command-line calculator.

Day 26

Learn

- logging

Add

Logs to expense tracker.

Day 27

Learn

- classes and OOP basics

Build

BankAccount class.

OOP is useful but secondary early on.

Day 28

Mini Project #3

Build

To-Do CLI App

Features:

- add tasks

- delete tasks
- persistent storage

Great progression project.

Day 29

Learn

- clean code principles
- naming
- function design

These practices improve maintainability.

Day 30

Review + GitHub Cleanup

Improve:

- README files
- folder organization
- comments
- screenshots

PHASE 3 — Python for Data + MLOps (Days 31–45)

Goal:
Move into data workflows and automation.

Day 31

Learn

- NumPy basics
- arrays
- vector operations

Day 32

Learn

- pandas DataFrames

Practice

Load CSVs.

Day 33

Learn

- filtering
- grouping
- aggregation

Build

Sales analyzer.

Day 34

Learn

- missing data handling

Practice

Dataset cleaning.

Day 35

Mini Project #4

Build

CSV Analytics Tool

Features:

- statistics
- summaries
- filtering

Very useful for MLOps prep.

Day 36

Learn

- matplotlib basics

Build

Simple charts.

Day 37

Learn

- environment variables
- .env files

Important for production workflows.

Day 38

Learn

- subprocess
- automation scripts

Build

Backup script.

Day 39

Learn

- pathlib deeper
- automation utilities

Build

File organizer.

Day 40

Learn

- HTTP basics
- APIs deeper

Build

Dataset downloader script.

Day 41

Learn

- testing basics
- pytest

Test

Existing functions.

Day 42

Learn

- debugging workflows

Read errors carefully and isolate problems.

Day 43

Mini Project #5

Build

Log Parser

Reads logs and extracts:

- errors
- warnings
- summaries

Very MLOps-relevant.

Day 44

Learn

- Makefiles basics
- automation workflows

Day 45

Review Day

Refactor projects using:

- smaller functions
- clearer names
- better structure

PHASE 4 — Intro MLOps Workflow (Days 46–60)

Goal:
Understand the shape of real ML workflows.

Day 46

Learn

- scikit-learn basics

Train a tiny model.

Day 47

Learn

- train/test split
- model evaluation

Day 48

Learn

- saving models
- pickle/joblib

Day 49

Build

Tiny ML training pipeline.

Structure:

src/
data/
models/

Day 50

Learn

- configuration files
- YAML basics

Day 51

Learn

- Docker basics

Containerize one app.

Day 52

Learn

- FastAPI basics

Create a simple API.

Day 53

Build

Model prediction API.

Day 54

Learn

- Git branches
- pull requests
- GitHub workflow

Day 55

Learn

- CI/CD concepts

- GitHub Actions basics

Day 56

Add

GitHub Actions to a repo.

Day 57

Build

End-to-end mini ML project.

Example:

- dataset
- training script
- saved model
- API endpoint

Day 58

Portfolio Polish

Improve:

- READMEs
- project descriptions
- architecture diagrams

Day 59

Resume + Portfolio Day

Create:

- pinned repositories
- portfolio README
- learning summary

Day 60

Final Review

Write:

What I Learned in 60 Days

- Python fundamentals
- APIs
- automation
- pandas
- ML basics
- Docker
- GitHub workflows

Then choose:

- deeper MLOps
- cloud
- orchestration
- Kubernetes
- Airflow
- ML pipelines

Recommended Daily GitHub Commit Types

Examples:

feat: added CSV parser
fix: corrected ValueError handling
refactor: improved function naming
docs: updated README

Best Resource Strategy

Use:

- documentation
- exercises
- building
- debugging

Avoid:

- passive binge-watching tutorials

Your Minimum Viable Portfolio by Day 60

You should ideally have:

Project	Level
Calculator	Beginner
Expense Tracker	Intermediate
To-Do App	Intermediate
CSV Analyzer	Intermediate

File Organizer	Intermediate
ML Pipeline Mini Project	MLOps Intro
FastAPI Prediction API	MLOps Intro

That is already a strong beginner portfolio.

Final Advice

Your competitive advantage will NOT be:

- knowing obscure syntax
- memorizing everything

It will be:

- consistency
- public documentation
- practical projects
- clean GitHub repos
- evidence of improvement over time

That combination stands out very quickly in junior MLOps paths.